
Hidden Consensus, Loud Disagreement: Autonomous ML Agents Diverge in What They Measure but Converge on What They Conclude

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Abstract

1 Independent AI agents given identical machine learning research questions and an
2 identical dataset (UCI Adult/Census Income) diverge sharply in what they choose
3 to report when a question’s operationalization is left unspecified — abstractly-
4 worded questions produced 9.1 unique measure framings per 10 replicates on
5 average across 120 independent, memory-isolated replicate analyses, versus 3.8
6 for concretely-specified questions. We show, however, that this surface-level di-
7 vergence substantially overstates genuine disagreement: on two abstract questions,
8 replicates converge on the same qualitative conclusion in 65–100% of cases despite
9 near-total disagreement on which specific number to headline — loud disagree-
10 ment about what to report can coexist with hidden consensus about what is true.
11 We additionally test a domain-appropriate intervention unavailable in prior (non-
12 executable) research settings — requiring agents to verify their own finding via
13 repeated cross-validation or bootstrap re-estimation before reporting — and find
14 it does not reduce measure-choice diversity, but appropriately tightens numeric
15 estimates for robust effects while surfacing genuine fragility in weak ones (sign
16 agreement on a small effect drops from 90% to 70% once agents are required to
17 check it). A matching 120-replicate dataset collected with Claude Haiku in place
18 of Sonnet replicates these patterns, but also reveals a further failure mode: on two
19 hypotheses where both model families converge almost completely on *what* to mea-
20 sure, they reach opposite conclusions about *what value they get* (one effect reverses
21 sign and grows $15\times$ larger), traced to an upstream preprocessing choice invisible
22 in the measure name itself — apparent consensus can mask real disagreement just
23 as easily as loud disagreement can mask real consensus. All reported findings were
24 independently verified via re-execution of each replicate’s saved analysis code
25 against the original data: 208/240 (87%) reproduced exactly or within tolerance.
26 The Sonnet dataset showed zero value mismatches; 4/120 Haiku replicates did,
27 traced to self-reported values not matching their own confirmed-deterministic code,
28 a distinct reliability gap we disclose in full.

29 **Keywords:** autonomous research, nonstandard errors, machine learning evaluation, multi-agent
30 verification, researcher degrees of freedom

31 1 Autonomous research result

32 This part is for the research itself: the hypothesis, method, evidence, and primary result generated or
33 developed by the agent.

1.1 Hypothesis

When independent human research teams are given the same dataset and question, they routinely reach different quantitative conclusions — not from error, but because the space of defensible analytical choices (model, controls, metric, validation scheme) is large, and different teams occupy different corners of it. Menkveld et al. [2024] documented this directly: 164 independent teams analyzing the same financial dataset produced estimates whose disagreement — “nonstandard errors” — often rivaled or exceeded each team’s own reported statistical uncertainty. Gao and Xiao [2026] recently showed autonomous AI agents reproduce this: 150 independent Claude Code agents given the same six market microstructure questions diverged sharply, driven by stable methodological forks, with AI peer review largely failing to resolve the disagreement while exemplar exposure did.

Both studies are confined to finance and social-science-style empirical analysis, where the “correct” measure is inherently a matter of judgment rather than something executable and checkable — unlike machine learning research, whose central operations (train a model, evaluate it, compute a metric) are directly checkable. This raises two questions neither prior study could address: does the same nonstandard-errors phenomenon appear when the domain provides mechanical ground truth, and, because ML tasks are executable, does asking agents to verify their own findings by re-running the analysis under a stability check reduce disagreement the way exemplar exposure did for Gao and Xiao — an intervention with no clean analogue where the underlying “truth” cannot itself be re-executed?

We hypothesize that (H-a) measure-choice divergence replicates in the ML domain, driven primarily by whether a question’s operationalization is specified, and (H-b) self-verification reduces numeric uncertainty in a chosen measure without resolving disagreement about which measure to choose. H-a predicts higher measure-framing diversity for abstract than concrete questions; H-b predicts the verification arm changes numeric dispersion/sign agreement without changing measure-choice diversity.

1.2 Method

Dataset. All replicates analyze a single, static copy of the UCI/OpenML Adult (Census Income) dataset (48,842 rows, mixed categorical/numeric features, binary income target, ~24% positive class), fetched once and frozen so every replicate receives byte-identical data.

Research questions. We wrote six research questions about this dataset, in three matched abstract/concrete pairs (Table 1). Each replicate agent receives only the dataset and one research question — no documentation, no access to any other replicate’s work, and no information about which other measures might be reasonable.

Table 1: The six research questions, in three abstract/concrete matched pairs.

ID	Specificity	Question
H1	Abstract	Does model family meaningfully affect predictive performance?
H2	Concrete	Does default-hyperparameter random forest beat default logistic regression on stratified 5-fold CV ROC-AUC?
H3	Abstract	Which features are most important for predicting income?
H4	Abstract	Does addressing class imbalance improve model quality?
H5	Concrete	Does SMOTE oversampling change minority-class F1 by more than 0.02 versus no resampling, holding the classifier fixed?
H6	Abstract	Is the model well-calibrated?

Isolation and scale. Every replicate runs as an independent, non-interactive Claude Code CLI invocation (`claude -p ... -dangerously-skip-permissions`) in a freshly created, unpredictably-named temporary directory with no persistent sibling directories visible at any point, and an explicit instruction not to explore outside its own working directory. We collected 10 independent replicates per (hypothesis $\times$ arm) cell — 120 replicates total with Claude Sonnet, plus a matching 120-replicate cross-model-family check with Claude Haiku (Section 2.3).

The verification arm. Half of the replicates for each hypothesis (the “verify” arm) receive an additional instruction: before reporting, validate your finding’s stability via repeated cross-validation with different seeds, a bootstrap confidence interval, or a held-out re-test split, and report whether it held up. The other half (“no-verify”) report directly after a single analysis pass.

76 **Structured reporting.** Every replicate writes a structured result (a named metric, its value, a direction,
77 and free-text on its methodological choices) rather than a forced common statistical form, since
78 abstract questions admit incompatible units.

79 **Verification of our own data.** Independent of the verification *arm*, we re-executed every replicate’s
80 saved code against fresh data: 208/240 (87%) reproduced. Sonnet showed zero contradicting
81 values; 4/120 Haiku replicates did, traced to self-reports not matching their own deterministic code
82 (Section 2.5).

83 1.3 Evidence

84 **Measure-choice diversity tracks abstractness.** Figure 1 shows the number of distinct measure
85 framings (collapsing trivial rewordings into one canonical form) each hypothesis produced across
86 20 replicates. The contrast is stark: H2 (concrete) produced only 2 distinct framings, essentially all
87 rewordings of “ROC-AUC difference (RF – LogReg)”; its abstract counterpart H1 produced 15. H5
88 (concrete) produced 10; H4 (abstract) produced 19. H3 and H6, abstract questions with an implicit
89 but unstated method, land in between (15 and 17): agents agreed on *method* (permutation importance;
90 10-bin ECE) while diverging on *target*. Collapsed into abstract vs. concrete, this is a $2.4\times$ difference
91 in mean unique framings (9.1 vs. 3.8 per 10) and $3.6\times$ in how much the modal framing dominates
92 (19% vs. 68%).

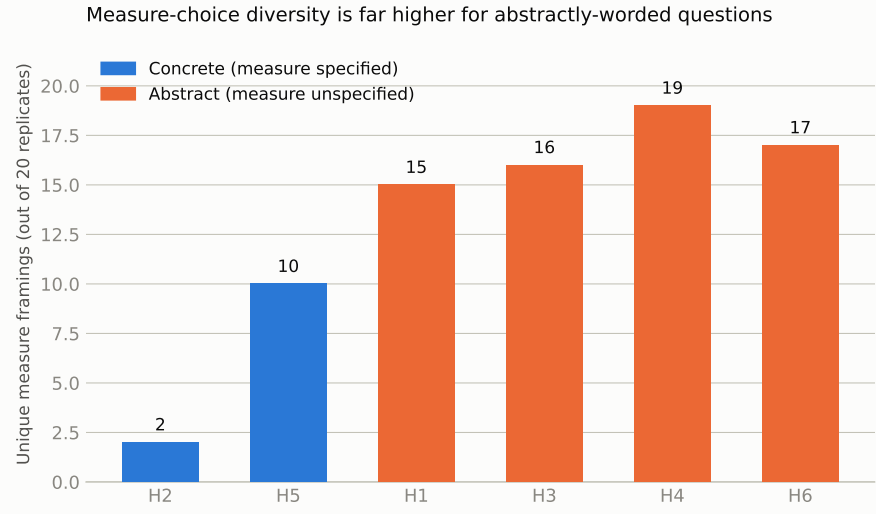


Figure 1: Unique measure framings per hypothesis (out of 20 replicates, 10 per arm). Concrete questions (blue) converge on a small number of framings; abstract questions (orange) produce near-total diversity.

93 **Surface disagreement overstates substantive disagreement.** The diversity counts above, read alone,
94 would suggest near-total disagreement on abstract questions — too strong a conclusion. For H1,
95 20/20 replicates (100%) agreed a tree-ensemble model (Gradient Boosting, HistGradientBoosting, or
96 Random Forest) outranks linear/naive baselines, despite zero agreement on which pair to headline.
97 For H4, 13/20 (65%) explicitly stated the full pattern “balanced accuracy/minority recall improves,
98 ROC-AUC stays roughly flat”; 18/20 (90%) agreed at minimum on “improves,” despite 19/20 unique
99 framings.

100 **Verification changes numeric confidence, not measure choice.** Figure 2 compares the no-verify
101 and verify arms for H2 and H5. For H2 — a large, robust effect — sign agreement is 100% in
102 both arms, and verification tightens the spread (SD 0.0038 $\rightarrow$ 0.0022). For H5 — small and fragile
103 — sign agreement *falls* under verification, 90% $\rightarrow$ 70%, and the mean effect shrinks toward zero
104 ($-0.0028 \rightarrow -0.0016$): several verify-arm replicates that saw an effect on one split found, once
105 checked via repeated CV, that the sign was unstable across folds, and reported this explicitly.

Verification tightens a robust effect (H2) and shrinks a fragile one toward zero (H5)

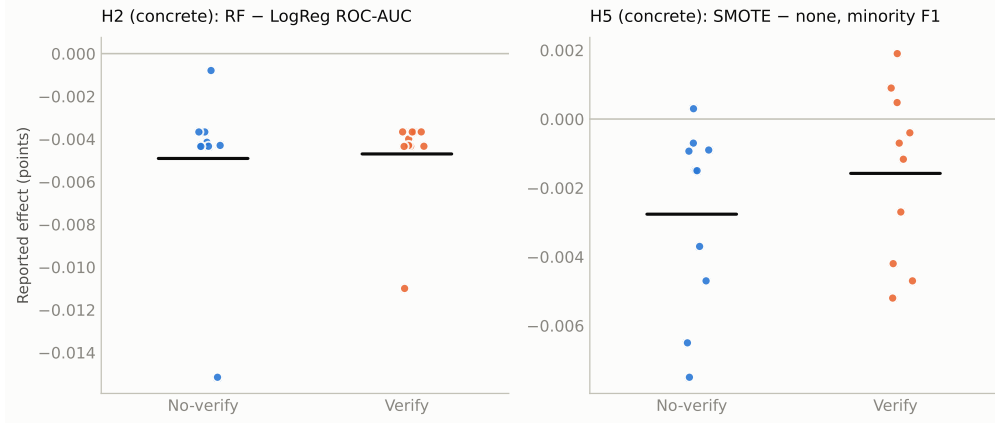


Figure 2: Replicate-level values for H2 and H5 by arm. Verification tightens a robust effect (H2) and shrinks a fragile one toward zero (H5); black bars show arm means.

A cross-model-family check surfaces hidden disagreement despite apparent consensus. The core patterns replicate with Haiku in place of Sonnet: unique framings remain higher for abstract than concrete questions (Table 2), and H1’s substantive convergence holds (18/20 Haiku vs. 20/20 Sonnet agree a tree ensemble wins) — though the gap’s size varies and even reverses for H6, where Haiku converged more (5 vs. 17). More strikingly, H2 and H5 — where both models agree almost completely on *what* to measure — reverse on *what value they get*: Sonnet unanimously finds logistic regression beating random forest on H2 (mean -0.0049); 90% of Haiku replicates find the opposite, $15\times$ larger (mean $+0.0765$), traced to Haiku disproportionately using naive label-encoding where Sonnet one-hot encoded — invisible in the measure name itself. This is close to the mirror image of our main finding: not loud disagreement masking hidden consensus, but apparent consensus masking real disagreement.

Table 2: Cross-model-family diversity: unique canonical measure framings per hypothesis (out of 20 replicates, both arms combined).

Hypothesis	Specificity	Sonnet	Haiku
H1	Abstract	15	17
H2	Concrete	2	1
H3	Abstract	16	19
H4	Abstract	19	20
H5	Concrete	10	15
H6	Abstract	17	5

1.4 Primary Result

Across six machine learning research questions and 240 independent, memory-isolated agent replicates (120 Sonnet, 120 Haiku): (1) question specificity is, as in prior non-ML domains, the primary driver of agent-to-agent disagreement, extending nonstandard errors in AI-conducted research [Gao and Xiao, 2026, Menkveld et al., 2024] into the one domain their method could not reach; (2) this surface-level divergence overstates genuine disagreement, with substantive conclusions converging in 65–100% of cases despite near-total disagreement on what to report; (3) self-verification recalibrates numeric confidence without touching measure-choice diversity, tightening robust effects and surfacing fragile ones; and (4) these patterns replicate across model families, but agreement on *what* to measure does not guarantee agreement on the value obtained — a second model family can reverse a unanimous finding entirely. Results (3)–(4) jointly show how “self-verification” should be read as a mitigation: it targets estimation noise in an already-chosen measure, with no mechanism to resolve disagreement about which measure was right, or what a different agent would have found.

2 System Design and Meta-Analysis

This is a human-author companion to the main result and should act as a meta-analysis of the research process undertaken.

2.1 Agent and Harness

All experimental replicates were produced by Claude Sonnet, invoked through the standalone Claude Code CLI (@anthropic-ai/claude-code, v2.1.221) rather than through the Anthropic API directly, using the author’s existing subscription rather than separate metered billing. Each replicate is a single, non-interactive invocation — `claude -p "<brief>" -dangerously-skip-permissions -model sonnet` — run with its working directory set to a freshly created, unpredictably-named temporary directory containing only a static copy of the dataset. `-dangerously-skip-permissions` was necessary for unattended batch execution across well over a hundred replicate runs, a real reduction in interactive safety guardrails relative to normal Claude Code use.

Isolation between replicates was approximated at two levels: a fresh, non-shared filesystem location per replicate, and an explicit brief instruction not to read, list, or explore any directory other than the replicate’s own working directory. This is weaker than the container/Singularity-based isolation used in the most directly comparable prior work [Gao and Xiao, 2026], which reports having observed agents reading sibling outputs without such isolation during their own trial runs; we did not build container-level isolation for this laptop-scale study (Section 2.8).

Every replicate was asked to produce a structured result file (a named primary metric, its value, a qualitative direction, and free-text methodological-choices text) plus its analysis code as a saved script, avoiding the unit-normalization problem Gao and Xiao [2026] had to solve after data collection by asking each agent to self-report one named metric and its own reasoning rather than forcing a common statistical form up front.

2.2 Tools

The agent’s own toolset within each replicate session was the standard Claude Code toolset (file read/write, bash execution) with no additional MCP servers or external tools configured; agents wrote and executed their own Python analysis code, predominantly using scikit-learn, pandas, and, for several replicates addressing class imbalance, imbalanced-learn/SMOTE. The orchestrating batch harness — dispatching replicates, managing isolation, collecting results, and the later verification and analysis passes — was written in Python using standard scientific-Python tooling (`concurrent.futures`, `pandas`, `scikit-learn`, `matplotlib`), run locally with no cloud infrastructure beyond the Claude Code CLI calls themselves. All experiments ran on a single Apple Silicon laptop with no dedicated GPU or compute cluster. Total data collection spanned roughly 33 calendar hours across five retry waves, but only about 3–4 of those hours involved any replicate actively running; the remainder was idle time between waves, reflecting both the CLI’s session usage limit and whenever the human author was next available to check and relaunch collection (Section 2.3).

2.3 Research Loop

The research loop had three phases: topic selection, data collection, and analysis.

Topic selection involved an extended, iterative novelty-checking process: roughly fifteen candidate research questions across three genres (LLM agent behavior/reliability; classical ML mechanisms; large-scale systematic ML audits) were proposed and checked against current literature via web search, and the great majority were found to already have close prior art, in several cases published within weeks of the check. The direction pursued here was arrived at only after an earlier framing was found to substantially overlap with Gao and Xiao [2026] and an adjacent cluster of very recent work, prompting a search for a structurally, not just incidentally, distinct angle: the ML domain those studies could not reach.

Data collection proceeded in waves rather than a single pass. During batch execution, the harness repeatedly encountered the Claude Code CLI’s session usage limit; since each replicate’s result is logged independently, the harness resumed by re-running only cells without a completed replicate,

so already-completed runs were never discarded. This required five separate retry waves spanning roughly 33 calendar hours, though only about 3–4 hours of that involved any replicate actively running — the rest was idle time between waves, reflecting both the session limit itself and when the human author was next available to check and relaunch collection. The practical per-window ceiling turned out to be far smaller and more variable (roughly 15–56 successful heavy-agentic runs per window) than anticipated. Collection also surfaced a genuine failure mode: a subset of replicates — disproportionately the more compute-heavy verification-arm ones — were found to spawn their analysis as a background shell task and then end their turn waiting for a completion notification that could never arrive in a single-shot, non-interactive session. An explicit instruction against this was added to the brief template; it measurably reduced but did not eliminate the failure mode. A separate lifecycle issue — three replicate slots that each succeeded twice under different run identifiers, traced to an orchestration process being halted without terminating its already-spawned subprocesses — was resolved by keeping the earlier-started run of each pair, a rule fixed before any outcome was inspected. The dataset was finally trimmed to a uniform $N=10$ replicates per cell (120 total) by restricting to a fixed replicate-index range in every cell, a rule fixed at dispatch time for presentational clarity.

Analysis proceeded through iterative drafting, figure generation, and review, detailed further in Section 2.4.

2.4 Human Interventions

The human author made every scope decision throughout the loop described in Section 2.3: rejecting an initial research direction once it was found to overlap with recent prior art and directing the agent toward the ML-domain extension; choosing the execution mechanism (CLI over a raw API key) and model-family scope (single family, deferred rather than abandoned); setting and later defending the replicate count ($N=10$ per cell) against a further reduction to $N=5$ that the agent had not proposed but the human considered and overruled on statistical grounds, since the verification-arm comparison — the paper’s most novel manipulation — would have had too little power at that sample size; and reviewing drafted figures and text at each stage. During review, the human author caught a specific framing problem the agent had not flagged on its own: an early draft title reused a precursor paper’s exact title as a substring, which would have read as a derivative reskin before a reader reached the abstract; the title was changed to lead with this paper’s own distinctive contribution instead. The human author also set the boundary that all reported figures (including replicate counts and the record of process incidents in this section) must reflect what actually happened, rather than a retrospectively cleaner narrative, and that this section’s substantial drafting collaboration with the agent be disclosed rather than presented as solely human-authored — consistent with, and enforcing, this workshop’s disclosure policy.

2.5 Verification

Two distinct verification mechanisms appear in this paper and should not be conflated. The **verification arm** is an experimental manipulation under study (Section 1.3), not a claim about data quality. Separately, we independently verified the **correctness of the underlying dataset** by re-executing every replicate’s saved analysis code against a fresh copy of the source data and comparing the reproduced primary metric to the one originally reported — a check made possible, and cheap, by the fact that ML analyses are directly executable, unlike the empirical-finance analyses in the most comparable prior work.

For the primary Sonnet dataset, 113 of 120 replicates (94%) reproduced exactly or within small numerical tolerance, and zero replicates showed a script running to completion and producing a value that contradicted its own report; every discrepancy traced to a limitation of the re-verification method itself (a timeout on the heaviest verification-arm scripts, and a smaller number of replicates whose saved script was not fully self-contained).

Running the same check on the Haiku dataset surfaced a different and more concerning pattern: 95 of 120 replicates (79%) reproduced, and this time 4 replicates’ saved scripts ran to completion and produced a value that contradicted their own report. We manually re-ran a sample of these mismatching scripts twice each and confirmed they are fully deterministic (identical output both times), ruling out simple run-to-run non-determinism; the discrepancy is instead between what the agent’s saved code actually computes and what it self-reported, which for these replicates

233 were apparently produced independently rather than the report being derived from that code’s own
234 execution. The remaining 21 non-reproducing Haiku replicates match the same not-self-contained-
235 script pattern seen in Sonnet, at a notably higher rate (17.5% vs. 5%).

236 We disclose all of this plainly: re-execution establishes that reported numbers were not fabricated on
237 the runs it could fully check and confirm, but it also surfaced — rather than merely failed to rule out —
238 a real, model-family-specific self-report reliability gap. This is itself consistent with this paper’s main
239 finding: verification is not one thing, and a check aimed at one failure mode (fabrication) can, run
240 rigorously enough, catch a different one (self-report/execution inconsistency) it was not specifically
241 designed to find.

242 2.6 Significance of result

243 The result matters for two different audiences. For researchers using AI agents as empirical collabo-
244 rators, it is a direct, actionable warning: an underspecified request to an AI agent does not fail loudly,
245 it fails quietly, by returning a confident, well-argued answer to a question subtly different from the
246 one intended — and the fix demonstrated here (specifying the measure, not just the goal) is cheap
247 and available today. For the autonomous-research community this workshop represents, the more
248 important result may be the second one: that a verification step targeting numerical stability does not,
249 and structurally cannot, resolve disagreement about what to measure in the first place. As autonomous
250 agents are given more latitude in real research pipelines, “have the agent check its own work” is a
251 natural first line of defense; this result suggests that defense has a specific, identifiable blind spot, and
252 that closing it likely requires a different kind of intervention (in the spirit of the exemplar-exposure
253 result of Gao and Xiao [2026]) rather than a more thorough version of the same one.

254 2.7 Interpretability

255 All experimental artifacts were tracked to support post-hoc audit. Every replicate’s raw result, saved
256 analysis code, and run metadata (timing, success/failure status, verification outcome) were written
257 to a dedicated, per-run directory identified by a unique run ID. Every batch dispatch and retry
258 wave was logged, one line per attempt, to an append-only log recording hypothesis, arm, replicate
259 index, and outcome, so the full history of successes, failures, retries, and duplicates is independently
260 reconstructable rather than overwritten. Excluded and superseded runs (failed attempts, and duplicate
261 successes resolved during data cleaning) were archived rather than deleted and remain available
262 for inspection. Separately from the experimental logs, a chronological process log recording every
263 scope decision, incident, and its resolution was maintained throughout the project and is available as
264 supplementary material; Section 2.3 and Section 2.4 are drawn directly from it.

265 2.8 Limitations

266 Isolation between replicates is prompt-level and filesystem-layout-based, not an OS-level sandbox
267 guarantee; we cannot rule out imperfect isolation in ways container-based isolation would have
268 prevented, though we have no direct evidence of cross-contamination in our data. This study covers
269 two model families (Sonnet, Haiku; Section 1.3); broader coverage (e.g. Opus, or non-Anthropropic
270 families) was not attempted, and unlike the primary Sonnet dataset, the Haiku dataset’s re-verification
271 (Section 2.5) was run as a follow-up check rather than alongside initial collection. N=10 replicates per
272 cell is well below the 150 used in the most comparable prior study, sized for laptop/subscription-quota
273 feasibility rather than by power analysis. Our re-verification of replicate outputs, while extensive,
274 could not fully check every replicate’s process end-to-end, for the reasons given in Section 2.5.

3 Reflections and Broader Impact

This paper’s main practical implication is a warning with an inverted risk profile from the usual failure mode people worry about with autonomous agents: the danger here is not that an agent visibly fails, but that it succeeds confidently and quietly at a slightly different task than intended, in a way that is only detectable by comparing many independent attempts. As autonomous agents take on a larger share of applied ML work — model selection, feature engineering, evaluation design — this failure mode will mostly go unnoticed by a single researcher running a single analysis, since nothing about a lone run signals that a different, equally defensible choice existed. The positive implication is equally direct: this risk is cheap to mitigate. Specifying the measure, not just the goal, collapsed measure-choice diversity by roughly $2.4\times$ in our data, at zero additional cost. We would encourage practitioners delegating open-ended ML analysis to agents to treat explicit measure specification as a default, not an afterthought.

For this workshop’s broader question of how the field should adapt, we would highlight two points beyond this paper’s specific result. First, our own novelty-checking process — roughly fifteen candidate directions checked against current literature, the great majority already covered by work published within weeks — suggests that “we searched and found nothing” is weakening as evidence of novelty specifically for autonomous research submissions, since the same tools that accelerate one group’s research loop accelerate everyone else’s. Venues may need to treat rapid, independent convergent discovery as an expected feature of the genre rather than a sign that a submission was insufficiently diligent. Second, self-verification is not one thing: this paper required two structurally different verification steps (an experimental manipulation testing whether agents checking their own numerical stability changes what they report, and an independent check that reported numbers were not fabricated), and conflating them would have been easy to do by accident. As “have the agent check its own work” becomes a standard mitigation for autonomous-agent unreliability, we would encourage reviewers and discussants to ask specifically which kind of wrongness a given verification step could and could not have caught, rather than treating any plausible-sounding check as covering all of them. Third, a laptop and a subscription are not yet a research lab, and the gap is measurable: this paper’s 120 replicates needed roughly 33 calendar hours and five retry waves despite only 3–4 hours of active computation (Section 2.3) — an artificial usage cap, not compute capability, where the most comparable prior study ran 150 agents against metered billing with no such wall [Gao and Xiao, 2026]. Access to a research lab increasingly means access to a *budget*, which shapes which questions are practically askable, not just how quickly they get answered — worth this workshop’s attention, since it bears on who gets to participate in autonomous research at all.

We do not see meaningful negative societal impact specific to this work: it uses a standard public benchmark dataset, introduces no new model or capability, and its practical recommendation (specify measures explicitly) reduces rather than increases risk from autonomous-agent-driven research.

A Reproducibility Statement

The dataset (UCI/OpenML Adult/Census Income) is public and freely available. The six research questions are given in full in Table 1; the exact brief text given to each replicate agent, including the verification-arm instruction, is available as supplementary material. Reproducing the replicate-collection process requires the Claude Code CLI and access to Claude Sonnet and Claude Haiku under the same or comparable versions; because model behavior can drift across versions, we do not expect bit-exact reproduction of individual replicate outputs, but do expect the qualitative pattern (higher measure-choice diversity for abstract questions; verification tightening robust effects and surfacing fragile ones) to be checkable by any lab with CLI access to a comparable agent. All analysis code (harness, verification, figure generation) is standard Python/scikit-learn and is included in supplementary material along with the full experimental log described in Section 2.3. *[Author note: confirm final code/data hosting location — e.g., an anonymized repository — before submission, and update this statement with the access details.]*

B Disclosure Statement

Agent(s) / model(s) used Claude Sonnet and Claude Haiku, via the Claude Code CLI (@anthropic-ai/claude-code v2.1.221). The agent conducted all 240 experimental replicates (120 Sonnet, 120 Haiku: data analysis, code writing and execution, result reporting) as described in Section 1–2, and substantially assisted in drafting this paper, including Section 1 in full and a first draft of Section 2, subsequently reviewed, corrected, and revised by the human author as described in Section 2.4.

☐ A human author has curated this work and verified every claim, figure, methodology detail, and reference.

☐ A human author takes full accountability for the submission as a human-authored, human-verified artifact.

[Author note: check both boxes only once you have personally verified every claim, figure, and reference below — do not check these before that review is actually done.]

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Justification: Section 2.8 discusses isolation guarantees (prompt-level, not OS-level), the two-model-family scope (broader coverage not attempted), sample size (N=10 per cell, well below the most comparable prior study's 150), and the documented limits of our re-execution verification method (Section 2.5).

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Answer: [Yes]

Justification: Section 1.2 fully specifies the dataset, the six research questions (Table 1), the isolation and replication design, the verification-arm manipulation, the model/CLI version, and the structured reporting format. Appendix A discusses what is needed to reproduce the qualitative pattern, noting that bit-exact reproduction of individual replicate outputs is not expected given possible model version drift over time.

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